HW 7, Math 104

Problem 1: A sequence of functions $\{f_n\}$, with $f_n : E \to \mathbb{R}$ ($E \subset \mathbb{R}$) is *pointwise bounded* if there there is a function $\phi(x) : E \to \mathbb{R}$ such that $|f_n(x)| < \phi(x)$ for all $x \in E$. The sequence is *uniformly bounded* if there is a number M such that $|f_n(x)| < M$ for all $x \in E$ and $n \in \mathbb{N}$.

a. Give an example of a sequence which is pointwise bounded but not uniformly bounded.

b. Prove that every uniformly convergent sequence of bounded functions is uniformly bounded.

c. Let $\{f_n\}$ be a sequence of continuous functions which converges uniformly to a function f on E. Prove that

$$\lim_{n\to\infty} f_n(x_n) = f(x)$$

for any sequence of points $x_n \in E$ such that $x_n \to x$.

d. Is the converse to (c) true?

Part A

Consider $f_n(x) = \frac{1}{nx}$ on $E = \{(0,\infty)\}$

Let $\phi(x) = \frac{2}{x}$

then $\frac{1}{nx} \leq \frac{1}{x} < \frac{2}{x}$ for $\forall x \in E$ and $\forall n \in \mathbb{N}$

then $|f_n(x)| < \phi(x)$ hence $\{f_n\}$ pointwise Bounded

Let $x = \frac{1}{M^2 n}$ for any number M on E

then $|f_n(x)| \geq M$ for all $x \in E$ and $n \in \mathbb{N}$

hence $\{f_n\}$ not uniformly bounded particularly in $(0,1)$

Part B

Let $\{f_n\}$ be a convergent Sequence in S

then for each $\varepsilon > 0$, there exists N such that $|f_n(x) - f(x)| < \varepsilon$ for $\forall x \in S$ and $\forall n > N$

Set $\varepsilon = 1$, by triangle inequality

$$|f_n(x)| < \varepsilon + |f(x)|$$
$$|f_n(x)| < 1 + |f(x)|$$

Set $M = \max\{|f(x)|+1, |f_1(x)|, |f_2(x)|, \dots, |f_N(x)|\}$ for $n \in \mathbb{N}$
hence, $|f_n(x)| < M$ for $\forall x \in S$ and $\forall n \in \mathbb{N}$
as desired, $\{f_n\}$ is uniformly bounded $\blacksquare$

<u>Part C</u>

Let $\{f_n\}$ be a sequence of continuous functions which converge uniformly to function f on E. Then
Let $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n > N$ implies

$$|f_n(x) - f(x)| < \frac{\varepsilon}{2} \quad \text{for } \forall x \in E$$

in particular, $|f_{N+1}(x) - f(x)| < \frac{\varepsilon}{2}$

Since f_{N+1} continuous at x_n by Theorem 4.3
let $\delta > 0$, then

$x \in E$ and $|x - x_n| < \delta$ implies
$$|f_{N+1}(x) - f_{N+1}(x_n)| < \varepsilon$$

So particularly for $n = N+1$
$$|f_n(x_n) - f(x)| \leq f_n(x_n) - f_n(x)| + |f_n(x) - f(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$
by triangle inequality hence $\lim_{n \to \infty} f_n(x_n) = f(x)$ $\blacksquare$

Part D

FALSE Consider the case where

We fix $x_n = a$ and $f_n(x) = x^n$ for $a = [0,1)$

then $f_n(x_n) = a^n \quad 50$,

$\lim_{n \to \infty} f_n(x_n) = 0$ for $x \in [0,1)$

f_n is pointwise convergent on $[0,1)$

for each $\varepsilon > 0$ and $x \in [0,1)$ there exists

N such that $|f_n(x_n) - 0| < \varepsilon$ for $n > N$

Because $f_n(a) = 0$ for some $a \in [0,1)$ and $\forall n > N$

But f_n is not uniformly convergent on $[0,1)$

that is we set $\varepsilon = 1/2$ then $\exists N$ such that

$|x_n - 0| < 1/2$ for all $x \in [0,1)$ & $\forall n > N$

Set $n = N+1$ and $x = (1/2)^{\frac{1}{N+1}} \in [0,1)$

then there is an $\varepsilon > 0$ such that this is not the case

$|f_n(x_n) - f(a)| < \varepsilon$ for all $x \in [0,1)$ & $\forall n > N$